

GSPS Implementation Brief

Single source of truth for the current pull request and Claude Code implementation

Scope: repository/PR discovery; qualifying-signal-to-plan automation; required entry confirmation; versioned indicator modules and PSAR alternatives; Wall Street Automation; paper/live isolation; live-only risk controls; backtesting and forward testing. Supersedes the prior implementation brief and all separate addenda.

Controlling directive

Implement a closed-loop Guided Stock Projection System (GSPS) in which a qualifying scan signal automatically creates one durable, idempotent candidate trade plan. A Wall Street-level member may deliberately activate a separate automation profile from the restricted Automation tab in paper or live mode. The scan itself must never activate live execution or submit an order.

The Seven Hermetic Principles, as discussed in Dennis William Hauck's *The Emerald Tablet: Alchemy for Personal Transformation*, are GSPS's product-design vocabulary and operating philosophy. They are not evidence that Hermetic concepts predict markets. Every strategy rule, parameter, eligibility decision, and performance claim must be explicit, versioned, reproducible, and empirically tested.

Repository and PR discovery

Claude Code must establish the actual repository, branch, pull request, implemented architecture, and exact missing files before it edits code. Do not infer or invent the project structure.

You are working in the current Git repository. Before implementing anything, identify the repository, current branch, and pull request.

- Run:

```
git remote -v
git branch --show-current
git status --short
git log --oneline -10
```
- If GitHub CLI is available and authenticated, run:

```
gh repo view --json nameWithOwner,url,defaultBranchRef
gh pr view --json number,title,url,body,state,baseRefName,headRefName,files,commits
```
- If `gh pr view` cannot identify a PR, inspect open PRs for the current head branch:

```
gh pr list --head "$(git branch --show-current)" --state open --json number,title,url,body,headRefName,baseRefName
```
- Search the repository and PR body/comments for:
 - "no UI"; "auto-creates a plan"; "qualifying signal"; "scan pipeline"
 - "trade plan"; "automation"; "paper trading"; "live trading"
 - "backtest"; "forward test"; "Wall Street"
 - "signal score"; "entry ready"; "confirmation move"; "retest"; "reclaim"; "reject"; "break sweep"
 - "PSAR"; "Parabolic SAR"; "Parabolic Stop"; "SetParabolicStop"; "acceleration factor"
- Search all code, migrations, routes, jobs, UI, configuration, and tests:

```
rg -n -i "qualif|signal|scan.*pipeline|trade.*plan|automation|paper.*trad|live.*trad|backtest|forward.*test|broker|order|position|s
```
- Read package.json/project scripts, migrations/types, API routes/server actions, scheduled jobs/queues, authentication/roles/RLS, etc.
- Before editing, write an evidence-based gap report with:
 - Repository and PR identified
 - Existing signal-to-plan flow
 - Existing signal-score and entry-confirmation flow
 - Existing PSAR/indicator usage and parameter persistence
 - Existing schema and migrations
 - Existing Automation UI/API
 - Existing paper/live separation
 - Existing broker, notification, and school integrations
 - Exact missing files and proposed files to change
 - Risks, assumptions, and true blockers

Do not merge, deploy, delete data, modify production secrets, create broker credentials, or submit real orders.

System layers

Layer	Required behavior	Authority
Signal	A deterministic, frozen GSPS strategy version evaluates a market condition	Scan/replay engine

Layer	Required behavior	Authority
Candidate plan	Each qualifying signal automatically produces one durable, idempotent candidate plan	State pipeline
Entry readiness	Candidate plan cannot become actionable until the defined break/retest/confirmation sequence completes	Simulation engine
Automation profile	Member deliberately elects to automate an eligible plan in paper or live mode	Wall Street member + server validation
Execution	Paper mode simulates. Live mode submits only server-authorized, GSPS-authorized orders.	Automation engine
Evidence	Backtest and forward tests use the same versioned signal and plan logic.	Research engine

Hermetic design mapping

Principle	GSPS expression
Mentalism	Explicit strategy rules, eligibility criteria, and immutable versioning precede discretionary interpretation.
Correspondence	Persist multi-timeframe, ticker/sector/index, instrument, and market-regime context.
Vibration	Measure momentum, volatility, volume, price movement, and market-data freshness.
Polarity	Every thesis has defined invalidation and opposite-direction logic; no automatic reversal without a newly qualified opposite plan.
Rhythm	Treat sessions, cycles, regimes, and holding periods as testable features rather than assumptions.
Cause and Effect	Persist the complete signal → plan → entry confirmation → order/simulation → outcome chain.
Gender	Generate candidates, then constrain activation and execution with validation, risk gates, and evidence.

End-to-end workflow

```

Market data or historical replay
→ scan run
→ normalized signal
→ qualification under frozen strategy + module version
→ qualifying signal persisted
→ candidate plan auto-created once (idempotent)
→ plan status: awaiting_entry_confirmation
→ break/sweep or breakout observed
→ return/retest/reclaim/rejection observed
→ confirmation move validated
→ entry_confirmed
→ Wall Street member activates paper or live automation profile
→ server pre-trade authorization
→ simulated order OR broker order
→ fills, positions, exit/loss events, audit trail
→ backtest and forward-test evidence

```

Automatic candidate-plan creation

A qualifying signal must create a candidate plan even if entry confirmation has not yet occurred. The plan remains non-executable until `entry\_confirmed`. A non-qualifying signal must create no plan. The system must never create duplicate plans when a scan repeats the same stable signal.

```

function candidatePlanKey(signal, strategy) {
  return [signal.ticker, signal.timeframe, strategy.version, signal.fingerprint].join(':');
}

if (signal.qualifies) {
  createOrUpsertCandidatePlan({
    sourceSignalId: signal.id,
    sourceScanRunId: scanRun.id,
    strategyVersion: strategy.version,
    strategyModuleVersion: module.version,
    idempotencyKey: candidatePlanKey(signal, strategy),
    generatedBy: 'gsps_scan_pipeline',
    status: 'awaiting_entry_confirmation',
  });
}

```

- Enforce the idempotency key with a database unique constraint/index.
- Persist source signal, scan run, data timestamp, current price, Gann decimal strip/digit sum/modulo-9 root, 3/6/9 harmonic vectors, entry zone, TP1, master target, stop/invalidation, polarity pivot, module configuration snapshot, strategy version, and audit timestamps.

- Record `PLAN\_AUTO\_CREATED\_FROM\_QUALIFYING\_SIGNAL` in immutable events.
- For a materially changed thesis, create a new/superseding plan instead of silently rewriting historical entry, stop, targets, or qualification evidence.
- Use fixed decimal/numeric monetary storage and calculations; never use JavaScript floating-point for money or loss thresholds.

Mandatory entry confirmation

GSPS recognizes that price commonly breaks above or below an entry level before returning. That initial move is a setup-state observation—not an executable entry. This rule applies to user-facing trade suggestions, backtesting, forward testing, paper orders, and live automated orders.

```
Signal lifecycle:
watching
  → setup_qualified
  → entry_zone_touched
  → break_or_sweep_detected
  → return_or_retest_detected
  → confirmation_pending
  → entry_confirmed
  → executed | expired | invalidated
```

An entry is allowed only after the defined return/retest/reclaim/rejection and confirmation move.

Score/gate component	Purpose	Can allow order?
Directional/Gann bias	Establishes bullish, bearish, or neutral strategic context.	No
Entry-zone proximity	Price reaches or initially crosses the GSPS zone.	No
Break/sweep or breakout	Price moves beyond the relevant level under configured rules.	No
Return/retest state	Price returns, retests, reclaims, or rejects the relevant zone.	No
Confirmation move	Approved close/structure/momentum/volume confirmation validates the return.	Potentially
Risk/freshness validation	Stop, targets, liquidity, data freshness, expiry, and risk/reward all pass.	Required
entryReady	Hard server-side final gate.	Yes

```
const entryReady =
  signalScore.total >= strategy.minimumSignalScore &&
  signal.breakOrSweepDetected &&
  signal.returnToEntryZoneConfirmed &&
  signal.confirmationMoveValidated &&
  signal.riskFeasible &&
  signal.isFresh &&
  !signal.isExpired &&
  !signal.isInvalidated;

if (!entryReady) {
  // A plan may display as forming/awaiting confirmation.
  // It must not create an actionable recommendation or order intent.
}
```

- For longs, define whether the required event is a sweep below and reclaim of the entry/support zone, a breakout-retest-hold, or another versioned setup rule.
- For shorts, define whether the required event is a sweep above and rejection below resistance/entry, a breakdown-retest-fail, or another versioned setup rule.
- A raw signal score never substitutes for a hard `entryReady` gate.
- An indicator flip, initial touch, initial break, or initial sweep alone cannot produce an entry or automatic reversal.
- Store timestamps, observed prices, and evidence for every state transition.

Indicator modules and PSAR

PSAR may remain as an optional research benchmark, but not the universal signal, entry, or reversal engine. Implement a data-driven, versioned GSPS indicator-module architecture. Each module must be constrained by GSPS qualification, instrument eligibility, risk rules, and test evidence.

Module	Primary purpose	Core components
GSPS Trend Rail	Default trend continuation and runner management.	Gann plan + entry zone + EMA structure + ADX regime filter + ATR Chandelier Exit.
GSPS Breakout Rail	Validated breakout trends.	Gann plan + Donchian/high-low breakout + ADX + ATR-based risk/trail.
GSPS Pullback Continuation	Trend pullback/retest entries.	Gann entry band + EMA pullback + price structure + RSI reset + volatility-aware risk.
GSPS VWAP Momentum	Liquid intraday equities/index futures.	Gann intraday levels + VWAP reclaim/rejection + relative volume + ATR risk trail.
GSPS Reversal Confirmation	Polarity shifts only after evidence.	Gann invalidation/pivot + market-structure break + RSI or MACD confirmation + ATR.
PSAR benchmark	Comparative research only.	Persist settings, calculation outputs, and results against GSPS alternatives.

- Inventory every existing PSAR use before changing it: location, entry/trend/trailing/reversal/display function, parameters, asset classes/timeframes, persistence, and parity across scan/backtest/forward/paper/live.
- Persist module name, module version, inputs, parameter snapshot, timeframe, calculation timestamp, market regime, and resulting evidence with signals and plans.
- Do not let a PSAR flip alone create an order, reverse a position, or bypass GSPS qualification/risk controls.
- A reversal requires plan invalidation or a formal GSPS polarity pivot plus a separately qualifying opposite-direction candidate plan.
- Compare PSAR and approved alternatives with identical symbols, periods, trading costs, fills, metrics, and out-of-sample/forward testing before live-automation approval.

Market-specific constraints

Market	Required GSPS treatment
Equities	EMA/ADX trend context; relative volume, sector/index context, and VWAP where intraday; validated percent/risk targets and run.
Options	Generate signal from the underlying-market GSPS plan; additionally gate liquidity, spread, expiration, Greeks, implied volatility.
Futures	Use session-aware rules, tick-value sizing, margin limits, contract liquidity, event windows, and ATR in ticks/dollars.
Forex	Use pip sizing, session overlap, spread, rollover, correlation, and economic-event gates.
Crypto/commodities	Apply liquidity/venue or contract-roll constraints, volatility regime controls, and market-specific risk units.

Automation tab

The Automation tab is reserved for Wall Street-level members. Server/API and RLS authorization must enforce this; a hidden UI route is not a security boundary.

Mode	Allowed behavior
System-plan automation	Member selects an eligible system-generated plan. All execution terms remain inside its GSPS rule envelope.
Guided custom automation	Member filters/selects eligible GSPS plans and configures only schema-permitted settings: allowed allocation, approval.
Paper	Creates simulated orders and positions only. It never calls a broker or triggers live-only consequences.
Live	May submit a broker order only after server-side entry confirmation, plan eligibility, risk validation, and execution auth.

- The scan auto-creates the candidate plan. The member—not the scan—chooses to activate paper or live automation.
- Use a separate `automation\_profiles` entity so multiple members can configure their own allocation, mode, broker connection, and risk limits against one candidate plan.
- The client submits an `automation\_profile\_id`; the server resolves all ticker, side, quantity, price constraints, stop/target logic, and broker intent. Never accept arbitrary order terms from the client.
- Automation profile execution mode is immutable after activation. A live position cannot be switched to paper to evade controls.

Live-only risk policy

The following requirements apply only when `execution\_mode = live`. They do not apply to paper trading. Paper trading may simulate outcomes but must not submit broker orders, issue mandatory live-risk notices, force close, restrict live access, or trigger a school re-completion requirement.

Live event	Required behavior
Default stop	Use the GPS-calculated stop/invalidation coordinate.
Widen/remove stop	Wall Street member may widen or remove the stop only after high-friction warning, verified email delivery, verified phone call.
6%, 9%, 15% trade loss	Send email and phone/SMS notification once per threshold per trade.
30% trade loss	Send hard warning regardless of membership level.
50% trade loss	Pause automation; cancel pending orders; submit/retry/reconcile close; restrict live trading; preserve paper access; record audit trail.

Loss thresholds are based on total funds allocated to the individual trade. Implement and test treatment for realized partial exits, remaining liquidation value, fees, and slippage. A forced close is an attempt and reconciliation process; record requested close time, order/fill/rejection data, and residual exposure because exact fills cannot be guaranteed in all markets.

```
if (position.executionMode !== 'live') return;

lossPct = ((allocatedTradeFunds - currentTradeLiquidationValue) / allocatedTradeFunds) * 100;

6%, 9%, 15%: email + SMS once each per trade
30%: hard warning once per trade, regardless of level
50%: pause profile → cancel pending orders → submit/reconcile flatten order
    → restrict live entitlement → allow paper → require school completion/re-completion
```

Pre-trade authorization

`authorizeAutomatedOrder(profileId)` resolves all order terms server-side.

Allow only when:

- profile mode is valid and immutable
- user has Wall Street entitlement
- plan is entry_confirmed, eligible, current, unexpired, un superseded, and not invalidated
- strategy/module version is approved for the market/instrument/timeframe
- data is fresh and liquidity/session gates pass
- configuration validates against the plan's GPS schema
- position/account/portfolio/daily risk controls permit it
- no global, member, or plan kill switch is active
- duplicate-order protection passes
- live mode: broker connection is valid, live access is not restricted, and stop-override/contact checks pass when required

All decisions, blocks, order intents, broker responses, fills, and lifecycle transitions are appended to an immutable audit ledger.

Backtesting and forward testing

- Backtesting must invoke the identical frozen strategy/module, qualification, candidate-plan, and ``entryReady`` logic that scan/paper/live automation uses. Use only information available at the decision timestamp and conservative fill/cost assumptions.
- Prevent look-ahead bias, survivorship bias, duplicate-signal inflation, and hidden parameter changes. Persist data source/version, dates, parameters, modeled costs/slippage, trade-level lifecycle evidence, and reproducible run identifiers.
- Forward testing creates timestamped candidate plans before their outcome is known. Original entry, stop, target, and configuration snapshots are immutable; new theses create new/superseding plans.
- Report expectancy, win rate, average win/loss, profit factor, maximum drawdown, count, time-in-trade, and breakdowns by strategy/module version, asset, timeframe, direction, root/vector, regime, and session.
- No module becomes eligible for live automation without an explicit approved state and recorded historical plus forward-test evidence.

Required UI

- Scanner/Trade Plans: scan health, freshness/errors, qualifying signals, candidate/active/invalidated/history states, and visible entry-confirmation status.
- Plan grid: ticker, direction, price, Gann root, entry zone, TP1, master target, stop, timeframe, status, age, source timestamp, and ``awaiting confirmation`` or ``entry confirmed`` state.
- Plan detail: full Gann Coordinate Execution Matrix; decimal-strip inputs; qualification rationale; indicator-module evidence; retest/confirmation timeline; polarity pivot; strategy/module version; source signal/run; and complete event history.
- Automation: eligible-plan list, block reasons, paper/live selection, bounded configuration form, risk preview, activation, personal and plan-level pause/stop, active state, and broker/order audit trail.
- Risk center: exposure, daily capacity, positions, P&L, broker/data health, notifications, restrictions, and kill-switch state.

- Label auto-created plans: “System-generated candidate — not an execution instruction.”

Implementation sequence

- Perform discovery and provide the gap report before edits.
- Preserve working logic while formalizing frozen strategy/module, signal, and entry-confirmation schemas.
- Create migrations, types, RLS, and authorization for signals, plans, event ledgers, automation profiles, execution data, risk controls, contact verification, restrictions, and school completion.
- Implement idempotent qualifying-signal → candidate-plan creation.
- Implement explicit entry lifecycle and the server-authoritative `entryReady` gate.
- Inventory/refactor PSAR into a benchmark-capable module; implement approved GSPS module architecture and calculation parity.
- Implement plan eligibility, bounded configuration validation, paper simulation isolation, and automation lifecycle.
- Implement live broker adapter behind pre-trade authorization, reconciliation, duplicate protection, and kill switches.
- Implement live-only stop override, notifications, loss monitor, forced-close/reconciliation, restriction, and school-gated reactivation.
- Implement Scanner, plan detail, Automation, and risk-center UI.
- Implement/extend backtest and forward-test evidence using the same plan engine.
- Run validation, document results, and update the existing PR. Do not merge or deploy.

Acceptance tests

- Non-qualifying signals create no candidate plan; qualifying signals create exactly one idempotent candidate plan.
- Candidate plans retain source, scan, strategy/module, Gann, entry, target, stop, polarity, and audit data.
- Initial entry-zone touch/break/sweep does not create an executable recommendation or order; the required return/retest plus confirmation does.
- Parity test: identical inputs produce the same score, lifecycle state, `entryReady` decision, and plan coordinates in scan, backtest, forward-test, paper, and live eligibility paths.
- PSAR flip alone cannot produce an entry/reversal; opposite-direction activity requires a separately qualifying GSPS plan.
- Only entitled Wall Street members can create/activate live automation; out-of-schema configuration is rejected server-side.
- Paper mode never calls a broker or causes live-only notices, forced closing, restrictions, or school re-completion requirements.
- Live stop widen/remove fails without entitlement, mandatory warning acknowledgement, verified email, verified phone/SMS, and audit events.
- Live alerts at 6%, 9%, 15%, and 30% occur once per trade.
- At live 50% loss, automation pauses, pending orders are cancelled, closure is attempted/reconciled, live access is restricted, and paper remains available.
- Restricted users cannot reactivate live automation until server-verified GSPS School completion/re-completion.
- UI covers loading, empty, blocked, error, and successful states and reads the server-backed audit trail.
- Lint, typecheck, tests, build, migration validation, and relevant security/RLS tests pass.

Claude Code implementation prompt

Implement this GSPS single-source-of-truth specification in the current repository.

FIRST: Complete the repository/PR discovery workflow and show the evidence-based gap report. Do not edit until that report is complete.

CORE: Preserve working qualification logic, but ensure every qualifying signal automatically persists exactly one idempotent GSPS candidate plan.

ENTRY CONFIRMATION: Price often breaks/sweeps beyond an entry level before returning. A touch, break, sweep, or indicator flip alone is not sufficient.

INDICATOR MODULES: Inventory all PSAR usage before modifications. Make strategy logic versioned and data-driven. Keep PSAR, if present, as a reference.

AUTOMATION: Implement/restrict a Wall Street-member Automation tab. Members deliberately create paper or live automation profiles from the plan.

PAPER: Simulation only. Never call a live broker and never apply mandatory live notices, forced close, live access restrictions, or school completion.

LIVE: A Wall Street-level member may retain, widen, or remove a stop only after an explicit high-friction warning, verified email delivery, and school completion.

EVIDENCE: Use the same frozen signal, plan, entry-confirmation, and strategy-module engine in scans, backtests, and forward tests. Pro

DELIVERABLES: Implement migrations/types/RLS, services, APIs/server actions, jobs, UI, tests, and documentation following existing pro

Non-negotiable constraints

- No claim that Hermetic principles establish market predictability; retain transparent empirical testing and risk disclosure.
- No client-side bypass of GSPS plan, entry-confirmation, entitlement, stop, risk, or execution restrictions.
- No arbitrary order endpoint accepting user-controlled ticker, side, price, stop, or quantity without server-derived authorization.
- No secrets or real broker credentials in source control.
- No merging, deployment, production changes, or real order submission as part of this task.
- No live-only enforcement applied to paper trading.